Detecting anomalous activity in a ship engine dataset

Analysis report

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Table of contents

Introduction

Ship engine failures can lead to increased operational costs, delayed deliveries, excessive fuel consumption, and safety risks. Detecting anomalous operating conditions before failures occur is therefore essential for predictive maintenance and efficient fleet management.

The objective of this project was to identify anomalous engine observations using both statistical and machine learning approaches. Three anomaly detection techniques were evaluated: the Interquartile Range (IQR), One-Class Support Vector Machine (OCSVM), and Isolation Forest (IF). Principal Component Analysis (PCA) was subsequently applied to visualise the machine learning results.

Methods

Dataset

The dataset used in this project contains sensor measurements collected from a ship's main engine. The dataset consists of 19,535 observations and six numerical features:

- Engine RPM (revolutions per minute)
- Lubrication oil pressure
- Fuel pressure
- Coolant pressure
- Lubrication oil temperature
- Coolant temperature

Each observation represents a snapshot of the engine's operating conditions at a specific point in time. No missing values or duplicated observations were found.

Exploratory Data Analysis

Descriptive statistics together with histograms, boxplots, violin plots and a correlation matrix were generated to understand the distributions of the variables and identify potential outliers.

The variables exhibited non-normal distributions, skewness and heavy tails. Boxplots confirmed the existence of extreme observations across all variables, although coolant temperature presented very few outliers compared with the remaining sensors. Correlation analysis indicated weak linear relationships between variables, suggesting that multivariate anomaly detection methods may provide additional information beyond independent feature analysis.

Anomaly Detection

The project specification required the identification of approximately 1–5% of observations as anomalous. This criterion was used as a stopping condition for each anomaly detection method. Three anomaly detection methods were applied independently to the dataset. A summary of each method is provided in Table 1.

Method	Type	Description	Parameters used
Interquartile Range (IQR)	Statistical	Observations are classified as anomalous when they fall below the lower bound *(Q1 − 1.5 × IQR)* or above the upper bound *(Q3 + 1.5 × IQR)*.	Number of simultaneous features containing anomalous values
One-Class Support Vector Machine (OCSVM)	Machine Learning	Learns the boundary of normal observations in a high-dimensional feature space and identifies points that lie outside this boundary as anomalies.	*nu, gamma*
Isolation Forest (IF)	Machine Learning	Builds an ensemble of random trees and isolates observations. Anomalies require fewer splits to isolate than normal observations and therefore have shorter average path lengths.	*contamination*

Table 1: Summary of anomaly detection methods used in this project.

A summary workflow of the analysis is presented in Figure 1. The OCSVM method is sensitive to differences in feature scales. Therefore, the dataset was standardised using the Standard Scaler method and used as an input to this method.

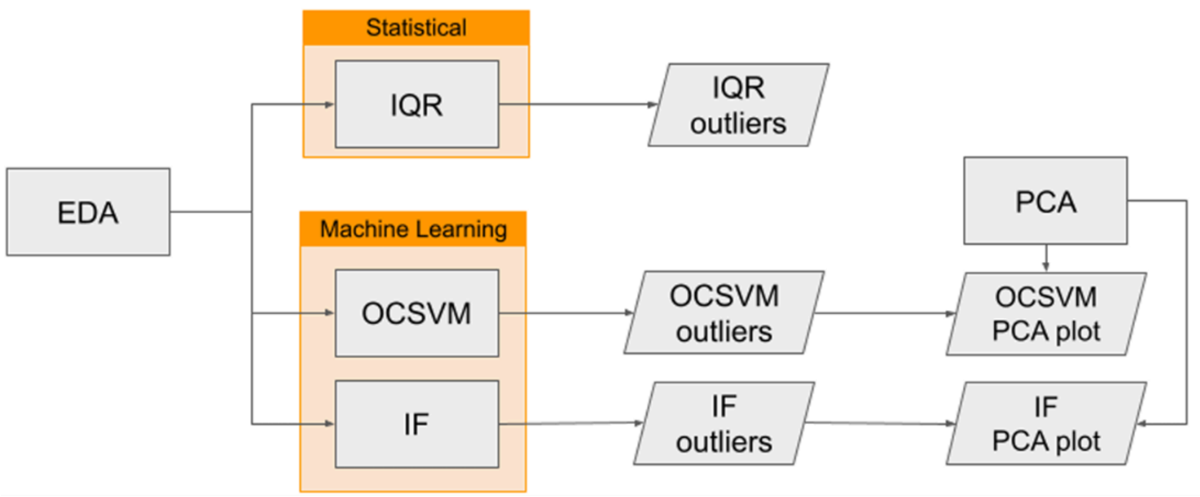

Figure 1: High-level overview of the analysis workflow used to perform anomaly detection in this project.

Principal Component Analysis (PCA)

After identifying the top 1–5% of anomalous observations using OCSVM and Isolation Forest, Principal Component Analysis (PCA) was performed to visualise the results. The scaled (standardised) dataset was used as an input to PCA. The first two principal components were used to create two-dimensional scatter plots in which observations Example project 4 identified as anomalies were highlighted. These visualisations were used to assess whether anomalous observations formed distinct patterns or clusters relative to normal observations.

Results and Discussion

EDA

Summary statistics for the ship engine dataset are presented in Table 2. No missing values or duplicate records were identified.

Statistic	Engine rpm	Lub oil pressure	Fuel pressure	Coolant pressure	Lub oil temp	Coolant temp
Count	19,535	19,535	19,535	19,535	19,535	19,535
Mean	791.2	3.3	6.7	2.3	77.6	78.4
Standard deviation	267.6	1.1	2.8	1.0	3.1	6.2
Minimum	61	0.0034	0.0032	0.0024	71.3	61.7
25th percentile (Q1)	593	2.5	4.9	1.6	75.7	73.9
Median (50%)	746	3.2	6.2	2.2	76.8	78.3
75th percentile (Q3)	934	4.1	7.7	2.8	78.07	82.9
Maximum	2239	7.3	21.1	7.5	89.6	195.5

Table 2: Summary statistics for the ship engine dataset.

The boxplots presented in Figure 2 indicate the presence of outliers in all six features. However, Coolant temperature contains only a small number of outlying observations, indicating that this variable is generally more stable than the others.

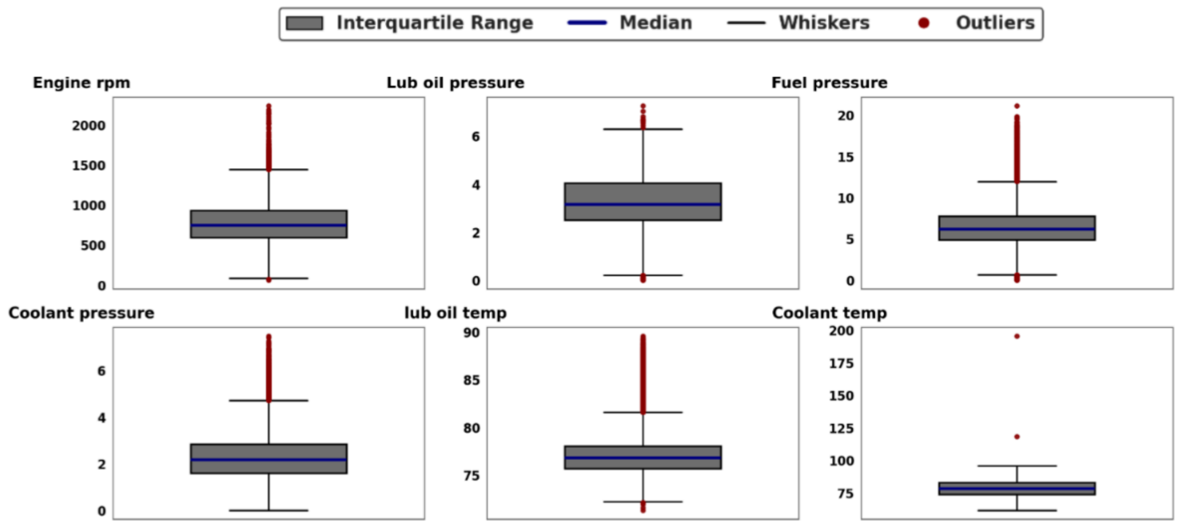

Figure 2: Boxplots for the six engine sensor features.

Interquartile Range (IQR)

Since anomalies occurring simultaneously across multiple engine measurements are more likely to represent genuine abnormal operating conditions, several thresholds based on the number of features exhibiting outlier behaviour were evaluated. The results are summarised in Table 3.

Simultaneous outlier features	Outlier count	Outlier percentage (%)
≥1	4,636	23.73
≥2	422	2.16
≥3	11	0.06
≥4	0	0.00
≥5	0	0.00

Table 3. Number of observations identified as anomalies using different IQR thresholds.

The results show that requiring **at least two simultaneous outlier features** is the only threshold that satisfies the expected anomaly rate of **1–5%**, identifying **422 observations (2.16%)** as anomalous. These observations were therefore selected for further analysis.

A pairwise co-occurrence analysis further showed that **lub oil temperature** most frequently appears together with anomalies in **fuel pressure** and **coolant pressure**, suggesting that abnormal lubricant temperature may be associated with broader changes in engine operating conditions.

One limitation of the IQR approach is that it evaluates each feature independently and therefore ignores the relationships between variables. Consequently, observations that appear normal when each sensor is analysed individually may still represent anomalous engine behaviour when the multivariate interactions among engine measurements are considered. This limitation motivates the use of machine learning approaches such as **One-Class SVM** and **Isolation Forest**, which are capable of detecting anomalies directly within the multidimensional feature space.

One-Class Support Vector Machine (OCSVM)

The OCSVM model was evaluated using different values of the **nu** and **gamma** parameters. The parameter **nu** controls the expected proportion of anomalies and therefore had a direct influence on the number of observations classified as anomalous. As expected, increasing **nu** produced progressively higher anomaly rates, ranging from **1.01%** to **5.00%**.

In contrast, varying the **gamma** parameter produced only minor changes in the results, indicating that the model is relatively insensitive to this parameter for the present dataset.

To facilitate comparison with the IQR analysis, the model with nu = 0.02 and gamma = "scale" was selected, as it identified 400 anomalous observations (2.05%), closely matching the 2.16% anomaly rate obtained using the IQR method. These observations were subsequently used for further analysis.

nu	**gamma**	**Anomalies detected**	**Percentage (%)**
0.01	scale	197	1.01
0.02	scale	400	2.05
0.03	scale	585	2.99
0.04	scale	785	4.02
0.05	scale	977	5.00
0.02	0.01	391	2.00
0.02	0.05	395	2.02
0.02	0.10	392	2.01
0.02	auto	400	2.05

Table 4: Number of anomalies detected by OCSVM for different nu and gamma values. The highlighted row shows the model chosen for further investigation.

Isolation Forest (IF)

The Isolation Forest model was evaluated using different values of the contamination parameter, which specifies the expected proportion of anomalies in the dataset. As shown in Table 5, the percentage of detected anomalies closely follows the selected contamination value, producing anomaly rates between 1.00% and 5.00%.

In addition, different numbers of trees (n_estimators) were evaluated. The results showed that increasing the number of trees did not modify the proportion of detected anomalies, indicating that the model is relatively stable with respect to this parameter.

To enable comparison with the IQR and OCSVM approaches, the model with contamination = 0.02 and n_estimators = 100 was selected, as it identified 391 anomalous observations (2.00%), closely matching the 2.16% anomaly rate obtained with IQR and the 2.05% obtained with OCSVM. These observations were subsequently used for further analysis.

Contamination	Anomalies detected	Percentage (%)
0.01	196	1.00
0.02	**391**	**2.00**
0.03	587	3.00
0.04	782	4.00
0.05	977	5.00

Table 5. Number of anomalies detected by Isolation Forest for different contamination values. The highlighted row corresponds to the selected model.

Principal Component Analysis (PCA)

The PCA projections for both **One-Class SVM** and **Isolation Forest** presented in **Figure 3** show similar patterns, with anomalous observations generally concentrated around the boundaries of the main data cloud but still exhibiting substantial overlap with normal observations. No clear separation between normal and anomalous operating conditions is visible in either projection.

Some observations identified as anomalous remain within the central cluster, while others located far from the majority of the data are classified as normal. This behaviour is expected because both OCSVM and Isolation Forest perform anomaly detection in the original six-dimensional feature space, whereas PCA provides only a two-dimensional representation of the data.

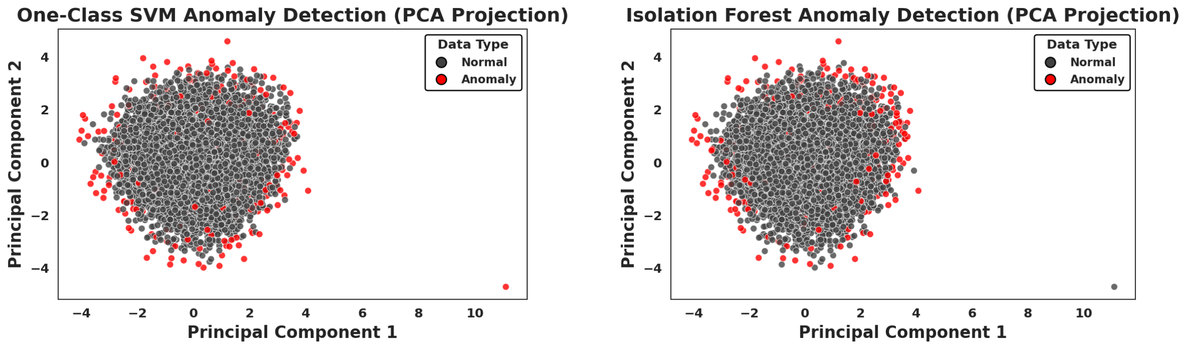

Figure 3. PCA projections for OCSVM (A) and Isolation Forest (B).

These findings suggest that the first two principal components do not fully capture the multidimensional structure used by the anomaly detection models. Consequently, observations that appear normal in the PCA projection may still exhibit anomalous behaviour when all six engine variables are considered simultaneously. Although PCA is useful for visualising the overall distribution of the data and identifying broad patterns, it should not be regarded as a definitive representation of anomaly separation in high-dimensional datasets.

Bonus: Engineering Interpretation and Practical Implications of the Detected Anomalies

Beyond Data-Driven Anomaly Detection

The anomaly detection techniques implemented throughout this project successfully identify observations that deviate from the normal operating behaviour of the ship engine dataset. However, these approaches remain fundamentally **data-driven**, meaning that they classify observations according to statistical or mathematical criteria without incorporating explicit engineering knowledge regarding marine diesel engine operation.

In industrial predictive maintenance, anomaly detection represents only the first stage of the diagnostic process. The true value of anomaly detection lies in understanding **why** a particular observation is abnormal and **which physical degradation mechanisms may be responsible**. Consequently, integrating domain expertise with machine learning models substantially improves interpretability and enables maintenance engineers to transform statistical anomalies into actionable maintenance decisions.

Recent studies in intelligent predictive maintenance highlight that hybrid frameworks combining artificial intelligence with engineering knowledge produce more reliable diagnostics than purely data-driven methods, particularly for complex thermodynamic systems such as marine diesel engines (Kalafatelis et al., 2025; Michel et al., 2026).

Engineering Relationships Between the Engine Variables

Marine diesel engines operate as highly interconnected systems in which changes occurring within one subsystem inevitably influence several other components. Therefore, simultaneous anomalies across multiple variables are generally more informative than isolated abnormal measurements.

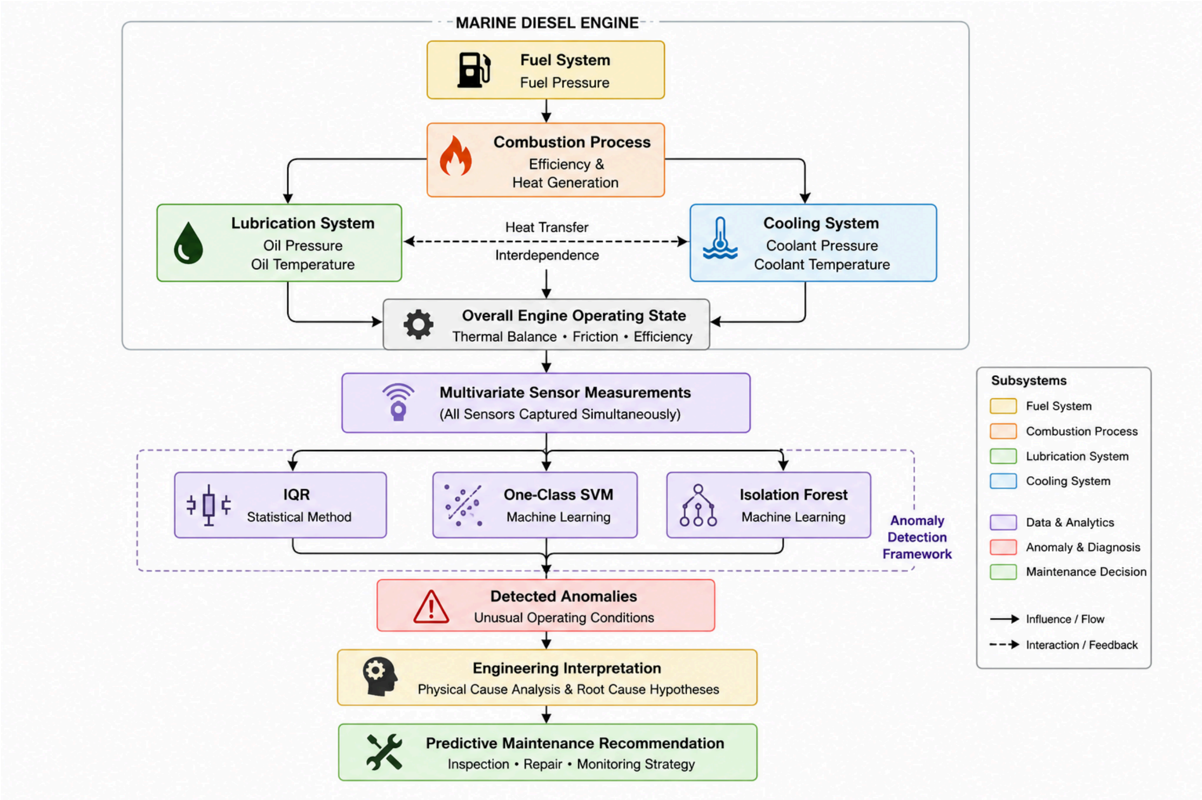

Figure 4. Interactions between marine diesel engine subsystems and the anomaly detection framework used in this study.

As illustrated above, the fuel injection system, cooling system and lubrication system continuously interact during engine operation. Consequently, deterioration affecting one subsystem often propagates through the remaining sensors, producing correlated anomalies rather than isolated abnormal observations.

Interpretation of the Detected Anomaly Patterns

The anomaly detection results indicate that **lubrication oil temperature frequently appears together with anomalies in fuel pressure and coolant pressure.** From an engineering perspective, this relationship is physically plausible.

An increase in lubrication oil temperature reduces lubricant viscosity and weakens the protective oil film separating moving mechanical components. This increases friction between bearings and rotating parts, producing additional heat generation throughout the engine. The excess thermal energy must subsequently be dissipated by the cooling system, potentially modifying coolant pressure behaviour and increasing thermal stress within the engine.

Similarly, abnormal fuel pressure may indicate injector fouling, fuel pump degradation or restrictions within the fuel delivery system. These faults may produce incomplete combustion and reduced combustion efficiency, generating irregular heat release inside the cylinders and increasing thermal loading on both the lubrication and cooling systems.

Therefore, simultaneous anomalies involving lubrication oil temperature, coolant pressure and fuel pressure may represent a common underlying degradation process rather than independent sensor abnormalities. This interpretation is consistent with established principles of marine engine diagnostics and predictive maintenance.

By contrast, coolant temperature exhibited relatively few anomalous observations. This behaviour is also physically reasonable, as marine diesel engines typically regulate coolant temperature through thermostatically controlled cooling circuits that maintain stable operating temperatures under varying engine loads. Pressure measurements generally respond more rapidly to hydraulic disturbances than temperature measurements, explaining why coolant pressure anomalies may occur while coolant temperature remains relatively stable.

Limitations of Purely Statistical Models

Although methods such as Interquartile Range (IQR), One-Class Support Vector Machine (OCSVM) and Isolation Forest successfully identify unusual observations, these algorithms cannot determine the physical cause of the detected anomalies.

The models operate exclusively within feature space and therefore detect deviations from the statistical distribution of the data rather than specific mechanical faults. Consequently, the presence of an anomaly should not be interpreted as direct evidence of component failure but rather as an indication that the observed operating conditions differ significantly from those previously encountered.

This limitation highlights the importance of combining machine learning techniques with engineering expertise. Knowledge-driven interpretation enables maintenance engineers to distinguish between harmless statistical variation and physically meaningful degradation mechanisms requiring intervention.

Practical Implications for Predictive Maintenance

The integration of anomaly detection with engineering knowledge allows maintenance actions to be prioritised according to the observed sensor combinations.

Detected sensor pattern	**Possible engineering interpretation**	**Recommended maintenance action**
High lubrication oil temperature	Reduced lubricant viscosity and increased friction	Inspect lubrication circuit and perform oil analysis
Low lubrication oil pressure	Oil pump degradation or internal leakage	Inspect pump, filters and lubrication lines

Abnormal fuel pressure	Injector fouling or fuel pump wear	Test fuel injection system and fuel filters
High coolant pressure	Cooling circuit restriction or pump malfunction	Inspect coolant pump, thermostat and heat exchanger
Simultaneous anomalies across multiple systems	Progressive engine degradation affecting multiple subsystems	Perform comprehensive predictive maintenance inspection

Table 6. Engineering interpretation of the detected anomaly patterns and recommended maintenance actions.

Rather than treating each anomaly independently, maintenance decisions should focus on combinations of abnormal variables that reflect the physical coupling between lubrication, combustion and cooling processes. Such an approach reduces unnecessary maintenance while increasing the probability of detecting genuine degradation before catastrophic engine failure occurs.

Industrial Relevance

From an industrial perspective, combining statistical anomaly detection with engineering interpretation transforms machine learning from a descriptive analytical tool into a practical decision-support system. Early identification of abnormal interactions between lubrication, cooling and fuel systems enables maintenance to be scheduled before severe efficiency losses, excessive fuel consumption or mechanical failures occur.

Consequently, integrating domain expertise with artificial intelligence contributes to improved operational reliability, reduced maintenance costs, lower vessel downtime and enhanced maritime safety. This hybrid methodology is increasingly recognised as the future direction of intelligent predictive maintenance systems for marine propulsion engines.

Conclusion

This project successfully applied statistical and machine learning techniques for anomaly detection in ship engine sensor data.

The IQR method provides a simple and interpretable baseline but cannot capture multivariate relationships between variables. OCSVM and Isolation Forest overcome this limitation by modelling the multidimensional feature space, allowing more complex abnormal operating conditions to be detected.

Among the evaluated methods, **Isolation Forest using contamination = 0.02 and 100 trees** provided the most robust solution due to its stable behaviour, low parameter sensitivity, computational efficiency, and anomaly rate consistent with the project requirements. OCSVM produced comparable results but requires feature scaling and is more sensitive to parameter selection.

Overall, combining statistical and machine learning approaches provides complementary information and yields a more comprehensive understanding of anomalous ship engine behaviour than any individual method alone.

References

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